# High-Order Geodesic Integration on Hessian Manifolds Without Third Derivatives: A Tutorial

**Abstract**

We describe a method for integrating geodesics on Hessian manifolds (the Riemannian geometry induced by the Hessian of a convex function) that achieves fourth-order accuracy using only gradient and Hessian evaluations—no third derivatives (Christoffel symbols) are needed. The method composes a second-order "Legendre midpoint" step using the classical Yoshida construction. We explain the ideas using only the geodesic equation and the Legendre transform, without the machinery of affine connections.

## 1 Setup

Let $F : \Omega \to \mathbb{R}$ be a strictly convex function on an open convex domain $\Omega \subset \mathbb{R}^n$. The *Hessian metric* is the Riemannian metric $g(s) = \nabla^2 F(s)$, which is positive definite by strict convexity.

The *geodesic* is the curve $\gamma(t)$ that locally minimizes the length

$$\ell(\gamma) = \int_0^1 \sqrt{\dot\gamma(t)^\top \nabla^2 F(\gamma(t))\,\dot\gamma(t)}\,dt.$$

The Euler–Lagrange equation for this variational problem gives the geodesic ODE:

$$\ddot\gamma^i + \frac{1}{2}\left[\nabla^2 F(\gamma)\right]^{-1}_{ik} \sum_{j,\ell} F_{j\ell k}(\gamma)\,\dot\gamma^j\dot\gamma^\ell = 0, \tag{1}$$

where $F_{j\ell k} = \partial^3 F/\partial s_j \partial s_\ell \partial s_k$. Integrating this ODE requires the *third derivatives* of $F$ (the "Christoffel symbols" of the metric).

**Goal.** We want to integrate (1) *without* computing $F_{j\ell k}$. We will use only:

- $\nabla F(s)$ (gradient), and
- $\nabla^2 F(s)$ (Hessian).

## 2 The Legendre transform and dual coordinates

The key tool is the *Legendre transform*. Define the dual coordinate

$$\lambda = -\nabla F(s).$$

This is a diffeomorphism from $\Omega$ to a dual domain $\Omega^*$ (the image of $-\nabla F$). Its inverse, which we write $s = \mathcal{L}(\lambda)$, satisfies $-\nabla F(\mathcal{L}(\lambda)) = \lambda$.

**Two coordinate systems.** The *same* Riemannian manifold $(\Omega, \nabla^2 F)$ can be described in two coordinate systems:

- **Primal coordinates** $s$: the original variables.
- **Dual coordinates** $\lambda = -\nabla F(s)$: the Legendre-transformed variables.

A straight line in primal coordinates ($s(t) = s_0 + tv$) is *not* a geodesic. A straight line in dual coordinates ($\lambda(t) = \lambda_0 + tw$) is also not a geodesic. But the true geodesic lies "between" these two approximations in a precise sense.

# 3 The Legendre midpoint step

The Legendre midpoint is a second-order approximation to the geodesic that uses one primal step and one dual step:

1. **Primal half-step.** Move halfway along a straight line in primal coordinates:
$$s_{1/2} = s_0 + \frac{h}{2}v.$$
2. **Evaluate dual at midpoint.** Compute the dual coordinate at $s_{1/2}$:
$$\lambda_{1/2} = -\nabla F(s_{1/2}).$$
Also let $\lambda_0 = -\nabla F(s_0)$.
3. **Dual extrapolation.** Take a full step in dual coordinates, using the midpoint velocity:
$$\lambda_1 = 2\lambda_{1/2} - \lambda_0.$$
(This is the midpoint rule: $\dot{\lambda} \approx (\lambda_{1/2} - \lambda_0)/(h/2)$, so $\lambda_1 = \lambda_0 + h \cdot \dot{\lambda}$.)
4. **Invert the Legendre transform.** Find $s_1$ such that $-\nabla F(s_1) = \lambda_1$. This is a nonlinear solve (Newton's method on $\nabla F(s) + \lambda_1 = 0$, using the Hessian $\nabla^2 F$ as the Jacobian).

## 3.1 Second-order accuracy

**Theorem 1** (Legendre midpoint matches the geodesic to second order)**.** *Let $F$ be a $C^3$ barrier on the interior of a proper cone, with Hessian $H_0 = \nabla^2 F(s_0)$. Let $\gamma$ be the geodesic from $s_0$ with velocity $v$ (i.e., $\gamma(0) = s_0$, $\dot{\gamma}(0) = v$, and $\gamma$ satisfies the geodesic equation* (1)*). Let $s_1$ be the output of the Legendre midpoint step with step size $h$. Then*
$$s_1 = \gamma(h) + O(h^3).$$
*More precisely: the Taylor expansions of $s_1$ and $\gamma(h)$ in powers of $h$ agree through order $h^2$, with*
$$s_1 = s_0 + hv + \frac{h^2}{2}\ddot{\gamma}(0) + O(h^3),$$
*where $\ddot{\gamma}(0) = -\frac{1}{2}H_0^{-1}T(v,v)$ and $T(v,v)_i = \sum_{jk} F_{ijk}(s_0)\, v_j v_k$.*

*Proof.* Write $H = H_0$, $\lambda_0 = -\nabla F(s_0)$, and $T(v) = T(v,v)$ for brevity. We compute the Taylor expansion of the Legendre midpoint output $s_1$ and show its $O(h^2)$ coefficient equals $\ddot{\gamma}(0)$.

Write $s_1 = s_0 + hv + \frac{h^2}{2}a + O(h^3)$ with unknown acceleration $a$. The proof determines $a$ by expanding each step of the algorithm.

*Step 1 (primal half-step).* $s_{1/2} = s_0 + \frac{h}{2}v$.

*Step 2 (dual at midpoint).* Taylor-expand $\nabla F$ at $s_{1/2}$:

$$\lambda_{1/2} = -\nabla F\big(s_0 + \tfrac{h}{2}v\big) = \lambda_0 - \tfrac{h}{2}Hv - \tfrac{h^2}{8}T(v) + O(h^3).$$

The third-derivative term $T(v)$ appears from the nonlinearity of $\nabla F$—this is where curvature enters.

*Step 3 (dual extrapolation).*

$$\lambda_1 = 2\lambda_{1/2} - \lambda_0 = \lambda_0 - hHv - \tfrac{h^2}{4}T(v) + O(h^3).$$

*Step 4 (inversion).* The algorithm defines $s_1$ by $-\nabla F(s_1) = \lambda_1$. We do not solve this equation; instead we determine the $O(h^2)$ coefficient $a$ by showing consistency.

Write $u = s_1 - s_0 = hv + \frac{h^2}{2}a + O(h^3)$ and Taylor-expand $\nabla F(s_0 + u)$ around $s_0$:

$$\nabla F(s_0 + u) = \nabla F(s_0) + Hu + \tfrac{1}{2}D^2(\nabla F)[u,u] + O(\|u\|^3).$$

The linear term gives $Hu = hHv + \frac{h^2}{2}Ha$. The quadratic term is $\frac{1}{2}T(u,u)$, where $T(u,u)_i = \sum_{jk} F_{ijk}u_j u_k$. Since $u = hv + O(h^2)$, we have $T(u,u) = h^2T(v,v) + O(h^3) = h^2T(v) + O(h^3)$. Combining (and using $\lambda_0 = -\nabla F(s_0)$):

$$-\nabla F(s_1) = \lambda_0 - hHv - \tfrac{h^2}{2}Ha - \tfrac{h^2}{2}T(v) + O(h^3).$$

*Matching.* Setting $-\nabla F(s_1) = \lambda_1$ and comparing the $O(h^2)$ coefficients from Step 3 and Step 4:

$$\underbrace{-\tfrac{1}{2}Ha - \tfrac{1}{2}T(v)}_{\text{from } -\nabla F(s_1)} = \underbrace{-\tfrac{1}{4}T(v)}_{\text{from } \lambda_1}.$$

Solving: $Ha = -\frac{1}{2}T(v)$, i.e., $a = -\frac{1}{2}H^{-1}T(v) = \ddot{\gamma}(0)$. □

**Where does the curvature come from?** The method never computes $F_{ijk}$ (third derivatives) explicitly. The curvature information is encoded in the *nonlinearity of* $\nabla F$: the Taylor expansion of $\nabla F(s_0+\frac{h}{2}v)$ contains the $\frac{h^2}{8}T(v)$ term automatically. The midpoint extrapolation (Step 3) amplifies it to $\frac{h^2}{4}T(v)$, and the inversion (Step 4) contributes another $\frac{h^2}{2}T(v)$. Together: $-\frac{1}{4} + \frac{1}{2} = \frac{1}{4}$, which after the $H^{-1}$ solve gives $\frac{1}{2}H^{-1}T$—exactly the geodesic acceleration.

**Analogy with mechanics.** This is the same principle as the Störmer–Verlet method: a half-step in position, a full step in momentum, a half-step in position. The Legendre transform $\lambda = -\nabla F(s)$ plays the role of $p = \partial L/\partial \dot{q}$. In both cases, the second-order accuracy comes from sampling the nonlinearity at the midpoint.

**Cost.** Each Legendre midpoint step requires:

- 2 gradient evaluations ($\nabla F$ at $s_0$ and $s_{1/2}$),
- 1 Newton solve ($\sim$3–5 Hessian evaluations for the inversion).

No third derivatives.

## 4 Exactness for the log barrier: Legendre midpoint = Padé

For the LP barrier $F(s) = -\sum_i \log s_i$, the Legendre midpoint is not merely a second-order approximation—it is *exactly* the [1/1] Padé approximant of the geodesic exponential.

**Theorem 2.** *Let $F(s) = -\sum_i \log s_i$ (the logarithmic barrier on $\mathbb{R}^n_+$). The geodesic from $s_0 > 0$ with velocity $v$ is $\gamma_i(h) = s_{0,i} \exp(hv_i/s_{0,i})$. The Legendre midpoint step from $s_0$ with step size $h$ and direction $v$ gives*

$$s_{1,i} = s_{0,i} \cdot \frac{1 + d_i/2}{1 - d_i/2}, \qquad d_i = \frac{hv_i}{s_{0,i}},$$

*which is the* [1/1] *Padé approximant of* $\exp(d_i)$*:*

$$\frac{1 + d/2}{1 - d/2} = \exp(d) + O(d^3).$$

*Proof.* The barrier has $\nabla F(s) = -s^{-1}$ (componentwise) and $\nabla^2 F(s) = \operatorname{diag}(s^{-2})$. The Legendre transform is $\lambda = -\nabla F(s) = s^{-1}$, with inverse $s = \lambda^{-1}$.

The four Legendre midpoint steps (componentwise, dropping the index $i$):

1. $s_{1/2} = s_0 + \frac{h}{2} v$.
2. $\lambda_{1/2} = s_{1/2}^{-1} = (s_0 + \frac{h}{2} v)^{-1}, \quad \lambda_0 = s_0^{-1}$.
3. $\lambda_1 = 2\lambda_{1/2} - \lambda_0 = \frac{2}{s_0 + \frac{h}{2}v} - \frac{1}{s_0} = \frac{2s_0 - (s_0 + \frac{h}{2}v)}{s_0(s_0 + \frac{h}{2}v)} = \frac{s_0 - \frac{h}{2}v}{s_0(s_0 + \frac{h}{2}v)}$.
4. $s_1 = \lambda_1^{-1} = \frac{s_0(s_0 + \frac{h}{2}v)}{s_0 - \frac{h}{2}v} = s_0 \cdot \frac{1 + d/2}{1 - d/2}, \quad$ where $d = hv/s_0$.

The [1/1] Padé approximant of $e^d$ is $(1 + d/2)/(1 - d/2)$, which satisfies $(1 + d/2)/(1 - d/2) = 1 + d + d^2/2 + \cdots = e^d + O(d^3)$. □

**Corollary 3.** *For $F(s) = -\log\det(S)$ on the PSD cone $\mathbb{S}^n_+$, the Legendre midpoint reduces to the* [1/1] *matrix Padé approximant of* $\exp(D)$ *when $S_0$ and $V$ are simultaneously diagonalizable.*

*Proof.* In the joint eigenbasis, $F$ decomposes into $-\sum_i \log \sigma_i$ where $\sigma_i$ are the eigenvalues. Each component follows Theorem 2 independently. □

**Remark.** The Yoshida fourth-order composition of three Padé [1/1] steps produces a higher-order rational approximation to $\exp(d)$—a [2/2]-like Padé obtained by composition rather than direct construction. For the log barrier, the Yoshida–Legendre method is therefore a *high-order rational approximation to the geodesic exponential* that generalizes seamlessly from symmetric cones (where it reduces to Padé) to arbitrary barriers (where it uses the Legendre transform in place of the matrix exponential).

## 5 The Yoshida fourth-order composition

The Legendre midpoint is second-order: the error is $O(h^3)$. To get fourth-order accuracy ($O(h^5)$ error), we use the *Yoshida construction* (1990).

**Idea.** If $\Psi_h$ is a second-order method (so $\Psi_h = \Phi_h + O(h^3)$ where $\Phi_h$ is the exact flow), then the composition

$$\Psi_h^{[4]} = \Psi_{x_1 h} \circ \Psi_{x_0 h} \circ \Psi_{x_1 h} \tag{2}$$

is fourth-order, provided

$$2x_1 + x_0 = 1 \quad \text{and} \quad 2x_1^3 + x_0^3 = 0.$$

The unique real solution is

$$x_1 = \frac{1}{2 - 2^{1/3}} \approx 1.3512, \qquad x_0 = \frac{-2^{1/3}}{2 - 2^{1/3}} \approx -1.7024.$$

**Why does this work?** The second-order method has error $\Psi_h = \Phi_h + c_3 h^3 + O(h^5)$ (no $h^4$ term because $\Psi$ is symmetric/time-reversible). The composition (2) gives

$$\Psi_h^{[4]} = \Phi_h + c_3(2x_1^3 + x_0^3)h^3 + O(h^5).$$

The condition $2x_1^3 + x_0^3 = 0$ kills the cubic error. The negative coefficient $x_0 < 0$ corresponds to a "backward" step that cancels the forward steps' third-order error.

**Applied to the Legendre midpoint.** The Yoshida fourth-order geodesic step is:

1. Legendre step with step size $x_1 \cdot h$ (forward),
2. Legendre step with step size $x_0 \cdot h$ (backward correction),
3. Legendre step with step size $x_1 \cdot h$ (forward).

Each Legendre midpoint step updates $(s, \lambda, v)$: position, dual, and velocity. The velocity at the end of each substep is recovered from the position change during the second primal half-step: $v_{\text{new}} = (s_{\text{new}} - s_{1/2})/(h_i/2)$.

**Cost.** Three Legendre steps $= 3 \times (2$ gradient $+1$ Newton$) = 6$ gradient evaluations $+3$ Newton solves. For a 3D cone (e.g., the exponential cone), this is $\sim$150 floating-point operations. No third derivatives at any point.

# 6 Summary of integrators

| **Method** | **Order** | **Needs $\partial^3 F$** | **Newton solves** | **Error at $\alpha$=0.1** |
|---|---|---|---|---|
| Euler | 1 | No | 0 | $2.3 \times 10^{-4}$ |
| Legendre midpoint | 2 | No | 1 | $7.7 \times 10^{-7}$ |
| Yoshida–Legendre | 4 | No | 3 | $2.5 \times 10^{-9}$ |
| Störmer–Verlet (8 steps) | 2 | Yes | 0 | $3.3 \times 10^{-7}$ |
| Primal-dual Verlet (8 steps) | 2+ | Yes | 1 | $3.1 \times 10^{-8}$ |

The Yoshida–Legendre method achieves fourth-order accuracy ($2.5 \times 10^{-9}$) without any third-derivative computation, outperforming even the primal-dual Verlet ($3.1 \times 10^{-8}$) which uses 8 substeps with full Christoffel symbol evaluations.

# 7 Why Yoshida and not Gauss–Legendre?

The Yoshida construction composes three [1/1] Padé steps to get fourth-order accuracy. A natural question: can we instead use the [2/2] Padé directly? For the log barrier, the [2/2] Padé of $\exp(d)$ has optimal error constants (about $40\times$ smaller than Yoshida at fifth order). The [2/2] Padé corresponds to the 2-stage Gauss–Legendre implicit Runge–Kutta method.

## 7.1 The obstruction

The Legendre midpoint works because the Legendre transform provides an *exact* dual evaluation at the midpoint: $\lambda_{1/2} = -\nabla F(s_{1/2})$. From the Taylor expansion proof (Section 3.1), the curvature enters through the nonlinearity of $\nabla F$ at the midpoint. The *symmetry* of the midpoint ($c = 1/2$) is essential: it is the unique point where the Strang splitting of primal and dual flat flows gives second-order accuracy.

The 2-stage Gauss–Legendre method evaluates at $c_1 = \frac{1}{2} - \frac{\sqrt{3}}{6} \approx 0.21$ and $c_2 = \frac{1}{2} + \frac{\sqrt{3}}{6} \approx 0.79$. At these non-symmetric points, we can evaluate $\lambda_i = -\nabla F(s_i)$ (the dual position), but we cannot recover the *velocity* $v_i$ at the stage points from the gradient map alone.

**The velocity problem.** At the midpoint ($c = 1/2$), the dual extrapolation $\lambda_1 = 2\lambda_{1/2} - \lambda_0$ is the *midpoint rule* for the integral $\lambda_1 = \lambda_0 + \int_0^h \dot{\lambda}\, dt$. The midpoint rule is second-order accurate: the error is $O(h^3)$ because the first-order error term vanishes when the sample point is at the center of the interval.

At a general GL node $c_i \neq 1/2$, the analogous extrapolation $\lambda_1 = \lambda_0 + (\lambda_{c_i} - \lambda_0)/c_i$ is *not* a midpoint rule—it is a forward-Euler extrapolation from an off-center sample point, which is only *first-order* accurate ($O(h^2)$ error). The GL consistency conditions couple $v_1$ and $v_2$ through the Butcher matrix $a_{ij}$, but without the geodesic acceleration (which requires $\partial^3 F$), the coupled system is underdetermined.

**What goes wrong.** We implemented the 2-stage GL with Legendre-derived velocities and fixed-point iteration. The result: the iteration does not converge to the geodesic. The error ($1.3\times10^{-4}$) is comparable to Euler, indicating that the Legendre velocity approximation destroys the fourth-order accuracy of the GL method.

**Why Yoshida avoids this.** The Yoshida construction sidesteps the velocity problem entirely: it composes three *midpoint* steps (where the Legendre trick is exact) with algebraic coefficients. The midpoint is the only node where the Legendre transform provides a second-order velocity without the ODE acceleration. Yoshida is the optimal way to achieve fourth order from this midpoint-only primitive.

| **Method** | **Order** | **Needs $\partial^3 F$** | **Error at $\alpha = 0.1$** |
|---|---|---|---|
| Legendre midpoint (= [1/1] Padé) | 2 | No | $7.7 \times 10^{-7}$ |
| Yoshida–Legendre (3 midpoint steps) | 4 | No | $2.5 \times 10^{-9}$ |
| Gauss–Legendre 2-stage (attempted) | 4 | **Yes***  | fails |
| [2/2] Padé (optimal, log barrier) | 4 | No† | $\sim 10^{-10}$ (theory) |

$^*$ Without $\partial^3 F$, the GL stage velocities are only first-order.
$^\dagger$ The [2/2] Padé is a closed-form rational function of $d$, specific to $F = -\log$. It does not generalize to arbitrary barriers.

**Summary.** The midpoint ($c = 1/2$) is special: it is the unique evaluation point where the dual extrapolation $\lambda_1 = \lambda_0 + (\lambda_c - \lambda_0)/c$ reduces to the *midpoint quadrature rule*, which is second-order accurate. At any other $c$, the extrapolation is first-order (forward Euler from an off-center point), and the resulting velocity is too inaccurate for a fourth-order method. Yoshida achieves fourth order by composing three midpoint evaluations—the only node where the Legendre transform gives second-order velocity—with algebraic weights that cancel the third-order error.

# 8 Connection to Hamiltonian mechanics

The structure exploited here is the same as in symplectic mechanics:

| **Hamiltonian mechanics** | **Hessian manifold** |
|---|---|
| Position $q$ | Primal coordinate $s$ |
| Momentum $p = \partial L/\partial \dot{q}$ | Dual coordinate $\lambda = -\nabla F(s)$ |
| Kinetic energy $T(p)$ | Primal "straight line" |
| Potential energy $V(q)$ | Dual "straight line" |
| Verlet = split $T + V$ | Legendre = split primal + dual |
| Yoshida on Verlet | Yoshida on Legendre |

In mechanics, the Störmer–Verlet splits the Hamiltonian $H = T(p) + V(q)$ into separately integrable kinetic and potential parts. This works because $T$ depends only on $p$ and $V$ depends only on $q$ ("separable").

On a Hessian manifold, the geodesic Hamiltonian $\mathcal{H}(s,p) = \frac{1}{2}p^\top[\nabla^2 F(s)]^{-1}p$ is *not* separable (the "kinetic energy" depends on $s$). But the Legendre transform provides a different splitting: primal coordinates (where straight lines are "free") and dual coordinates (where straight lines are also "free"). The Legendre midpoint splits along this primal-dual axis instead of the kinetic-potential axis.

The Yoshida construction is purely algebraic—it works for any symmetric second-order method, regardless of the underlying geometry. Applied to the Legendre midpoint, it gives fourth-order accuracy on any Hessian manifold, for any strictly convex $F$, without ever computing the curvature.

# 9 Error bounds from self-concordance

Self-concordance of the barrier provides *a priori* error bounds for all the integrators, in terms of a single scalar parameter $\delta = h\|v\|_H$ (the step size in the local Hessian norm).

## 9.1 Self-concordance inequality

A $\nu$-logarithmically homogeneous self-concordant barrier $F$ satisfies:

$$|D^3F(s)[u,v,w]| \leq 2\,\|u\|_H\|v\|_H\|w\|_H \tag{3}$$

for all $u, v, w$, where $\|u\|_H = \sqrt{u^\top \nabla^2 F(s)\, u}$ is the local Hessian norm. This is the polarized form of $|D^3F[v,v,v]| \leq 2\|v\|_H^3$.

### 9.2 Tangent line (1st order)

The tangent line $L(h) = s_0 + hv$ approximates the geodesic $\gamma(h) = s_0 + hv + \frac{h^2}{2}\ddot{s} + O(h^3)$. The leading error is

$$\gamma(h) - L(h) = -\frac{h^2}{4} H^{-1} T(v, v) + O(h^3),$$

where $T(v, v)_i = \sum_{jk} F_{ijk} v_j v_k$.

**Theorem 4** (Tangent line error bound)**.** *Let $F$ be a $\nu$-LHSCB with self-concordance parameter $M_f$ (so (3) becomes $|D^3F[u, v, w]| \le 2M_f \|u\|_H \|v\|_H \|w\|_H$). Let $\gamma$ be the geodesic from $s_0$ with velocity $v$, and set $\delta = h\|v\|_{H_0}$, $H_0 = \nabla^2 F(s_0)$. If $M_f \delta < \frac{1}{4}$, then*

$$\|\gamma(h) - (s_0 + hv)\|_{H_0} \le M_f\, \delta^2. \tag{4}$$

*Proof. Step 1: geodesic acceleration.* The geodesic equation gives $\ddot{\gamma} = -\frac{1}{2} H(\gamma)^{-1} T(\dot{\gamma}, \dot{\gamma})$, so $\|\ddot{\gamma}\|_{H(\gamma)} = \frac{1}{2}\|T(\dot{\gamma}, \dot{\gamma})\|_{H(\gamma)^{-1}}$. For any $w$ with $\|w\|_{H(\gamma)} = 1$:

$$|w^\top T(\dot{\gamma}, \dot{\gamma})| = |D^3F[\dot{\gamma}, \dot{\gamma}, w]| \le 2M_f \|\dot{\gamma}\|^2_{H(\gamma)}.$$

Since geodesics have constant speed $\|\dot{\gamma}(t)\|_{H(\gamma(t))} = \|v\|_{H_0}$, we obtain

$$\|\ddot{\gamma}(t)\|_{H(\gamma(t))} \le M_f \|v\|^2_{H_0} \tag{5}$$

for all $t$ in the domain of $\gamma$.

*Step 2: trajectory containment.* Set $\beta(t) = M_f \|\gamma(t) - s_0\|_{H_0}$. On any interval where $\beta < \frac{1}{2}$, the Hessian comparison theorem gives $(1-\beta)^2 H_0 \preceq H(\gamma) \preceq (1-\beta)^{-2} H_0$, so $\|w\|_{H_0} \le (1-\beta)^{-1}\|w\|_{H(\gamma)}$ and

$$\|\ddot{\gamma}(t)\|_{H_0} \le \frac{M_f \|v\|^2_{H_0}}{1 - \beta(t)} \le 2M_f \|v\|^2_{H_0}.$$

Integrating: $\|\dot{\gamma}(t) - v\|_{H_0} \le 2M_f \|v\|^2_{H_0}\, t$, and then

$$\beta(t) \le M_f \big(t\|v\|_{H_0} + M_f \|v\|^2_{H_0} t^2\big) \le M_f\delta + (M_f\delta)^2 < \tfrac{1}{4} + \tfrac{1}{16} = \tfrac{5}{16} < \tfrac{1}{2}.$$

Since $\beta(0) = 0$ and the bound shows $\beta$ cannot reach $\frac{1}{2}$ on $[0, h]$, the containment holds by continuity.

*Step 3: error bound.* With $\|\ddot{\gamma}(t)\|_{H_0} \le 2M_f \|v\|^2_{H_0}$ for all $t \in [0, h]$:

$$\|\gamma(h) - s_0 - hv\|_{H_0} = \left\| \int_0^h (h - t)\, \ddot{\gamma}(t)\, dt \right\|_{H_0} \le 2M_f \|v\|^2_{H_0} \cdot \frac{h^2}{2} = M_f\, \delta^2. \qquad \square$$

**Remark.** The constant $M_f$ (rather than $\frac{1}{2}M_f$) is the price of a uniform bound over the trajectory: the Hessian comparison factor $(1-\beta)^{-1} \le 2$ converts the instantaneous bound (5) to the base-point norm. For $M_f\delta \ll 1$, the true error approaches the leading-order value $\frac{1}{2}M_f\delta^2$. For $M_f = 1$, the bound gives $\delta^2$, and the tangent line stays in the Dikin ellipsoid whenever $\delta < 1$.

### 9.3 Legendre midpoint (2nd order)

From the Taylor expansion proof (Section 3.1), the Legendre midpoint matches the geodesic to second order. The leading error is $O(h^3)$ and involves $D^4F$ (fourth derivatives) contracted with $v$. For a self-concordant barrier, iterated application of (3) gives:

$$\|\gamma(h) - s_{\text{Leg}}\|_H \le C_2\, \delta^3 + O(\delta^4), \tag{6}$$

where $C_2$ is a universal constant (independent of the specific barrier) that depends only on the self-concordance parameter.

**Sketch.** The $O(h^3)$ error involves the fourth-order Taylor coefficient: $D^4F$ contracted with $v$ three times and $H^{-1}$. By the derivative of self-concordance (differentiating (3) with respect to $s$), the fourth derivative satisfies $|D^4F[u,v,v,v]| \le C\|u\|_H\|v\|_H^3$. The $H^{-1}$ contraction contributes another $\|v\|_H$ factor, giving $\delta^3 = h^3\|v\|_H^3$ overall.

### 9.4 Yoshida–Legendre (4th and 6th order)

The Yoshida composition cancels the $O(\delta^3)$ error, leaving:

$$\|\gamma(h) - s_{\text{Yoshida-4}}\|_H \le C_4\,\delta^5 + O(\delta^6), \tag{7}$$

$$\|\gamma(h) - s_{\text{Yoshida-6}}\|_H \le C_6\,\delta^7 + O(\delta^8). \tag{8}$$

The constants $C_4, C_6$ involve higher derivatives of the self-concordance condition but remain bounded by universal functions of $\nu$ (the barrier parameter).

### 9.5 Summary: interior guarantee

| **Method** | **Error bound** | **Order** | **Interior if** |
|---|---|---|---|
| Tangent line | $\delta^2/2$ | 1 | $\delta < \sqrt{2}$ |
| Legendre midpoint | $C_2\delta^3$ | 2 | $\delta < C_2^{-1/3}$ |
| Yoshida-4 | $C_4\delta^5$ | 4 | $\delta < C_4^{-1/5}$ |
| Yoshida-6 | $C_6\delta^7$ | 6 | $\delta < C_6^{-1/7}$ |

Higher-order methods allow larger steps $\delta$ while staying interior. The constants $C_k$ are universal (depending only on $\nu$, not on the specific cone), so the step-size bounds are *a priori*—computable before the step is taken.

**Practical implication.** For an IPM with step size $\alpha$ and direction $d$, the Hessian norm $\delta = \alpha\|d\|_H$ is already computed during the line search. The error bound guarantees that the geodesic step stays interior whenever $\delta$ is below the method-specific threshold. No post-hoc feasibility check is needed.

## 10 Error analysis for general $M_f$

The error bounds in Sections 3.1–**??** were stated for the standard self-concordant case $M_f = 1$. We now give rigorous bounds for general $(M_f, \nu)$-self-concordant barriers, showing how the integrator order controls the dependence on $M_f$.

We begin with a digression on a weaker regularity condition that captures what the integrators use in practice: they never evaluate $D^3F$, only $\nabla F$ and $\nabla^2 F$ at different points.

### 10.1 Relative smoothness

**Definition 1.** *The gradient* $\nabla F$ *is* $L$-smooth relative to $\nabla^2 F$ *if for all* $s \in \operatorname{int}(K)$ *and* $\|u\|_{H(s)} < 1/L$*:*

$$\|\nabla F(s+u) - \nabla F(s) - \nabla^2 F(s)\cdot u\|_{H^{-1}} \le \frac{L\,\|u\|_H^2}{1 - L\|u\|_H}, \tag{9}$$

*where $H = \nabla^2 F(s)$ and $\|\cdot\|_{H^{-1}}$ is the dual norm. The domain restriction $\|u\|_H < 1/L$ and the denominator $1 - L\|u\|_H$ are the Hessian-metric analogs of the Dikin ellipsoid.*

This is the scale-invariant, self-referential analog of classical $L$-smoothness ($\nabla^2 F \leq LI$):

| **Classical (Euclidean)** | **Self-referential (Hessian)** |
|---|---|
| $\nabla^2 F \leq L \cdot I$ (smooth) | $\nabla F$ varies $\leq L$ relative to $\nabla^2 F$ |
| $\nabla^2 F \geq \mu \cdot I$ (strongly convex) | trivially holds with $\mu = 1$ |
| Step size $\delta < 1/L$ (absolute) | Step size $\delta < 1/L$ in local Hessian norm |

**Relation to self-concordance.**

**Definition 2** (Generalized self-concordance [**?**, Def. 5.1.1])**.** *A $C^3$ convex function $F$ is $(M_f, \nu)$-*self-concordant *if for all $s \in \operatorname{dom} F$ and $v \in \mathbb{R}^n$,*

$$|D^3 F(s)[v, v, v]| \;\leq\; 2M_f \, \|v\|^3_{H(s)}, \tag{10}$$

*where $H(s) = \nabla^2 F(s)$. The standard self-concordant case is $M_f = 1$.*

**Proposition 1.** *If $F$ is $(M_f, \nu)$-self-concordant, then $\nabla F$ is $L$-smooth relative to $\nabla^2 F$ with $L = M_f$.*

*Proof.* Write the Taylor remainder as

$$\nabla F(s + u) - \nabla F(s) - H(s)\, u \;=\; \int_0^1 \big[H(s + tu) - H(s)\big]\, u \,\mathrm{d}t.$$

Set $\alpha = M_f \|u\|_{H(s)} < 1$. The Hessian comparison theorem for self-concordant functions [**?**, Thm. 5.1.1] gives

$$H(s + tu) \;\preceq\; \frac{1}{(1 - t\alpha)^2}\, H(s).$$

Therefore $H(s + tu) - H(s) \preceq \big[(1 - t\alpha)^{-2} - 1\big] H(s)$, and taking the $H(s)^{-1}$ norm of the integral:

$$\begin{aligned}
\big\|\nabla F(s{+}u) - \nabla F(s) - H(s)u\big\|_{H^{-1}} &\leq \|u\|_H \int_0^1 \Big(\frac{1}{(1 - t\alpha)^2} - 1\Big) \mathrm{d}t \\
&= \|u\|_H \left[ \frac{1}{\alpha(1 - t\alpha)} \bigg|_0^1 - 1 \right] \;=\; \|u\|_H \cdot \frac{\alpha}{1 - \alpha} \;=\; \frac{M_f \, \|u\|_H^2}{1 - M_f \|u\|_H},
\end{aligned}$$

which is (9) with $L = M_f$. $\square$

**The converse is false.** A barrier can be $L$-smooth relative to $\nabla^2 F$ (finite $L$) even when $D^3 F$ has localized spikes (large or undefined $M_f$). Relative smoothness is strictly weaker: it measures the *integrated* gradient deviation, not pointwise third derivatives. This gap is practically significant for approximate barriers (where $D^3 F$ may not exist everywhere) and composite barriers (where a small perturbation can create large pointwise $D^3 F$ without affecting the integrated bound).

### 10.2 Error bounds for self-concordant barriers

Throughout this section, $F$ is $(M_f, \nu)$-self-concordant, $\gamma$ is the geodesic from $s_0$ with velocity $v$, and $\delta = h\|v\|_{H_0}$ with $H_0 = \nabla^2 F(s_0)$.

### 10.2.1 Preliminaries

**Lemma 1** (Geodesic acceleration)**.** *Along the geodesic, the acceleration in the instantaneous metric satisfies*

$$\|\ddot{\gamma}(t)\|_{H(\gamma(t))} \le M_f \|v\|_{H_0}^2 \tag{11}$$

*for all $t$ in the domain of $\gamma$.*

*Proof.* The geodesic equation gives $H(\gamma)\ddot{\gamma} = -\frac{1}{2}T(\dot{\gamma},\dot{\gamma})$, so $\|\ddot{\gamma}\|_H = \frac{1}{2}\|T(\dot{\gamma},\dot{\gamma})\|_{H^{-1}}$. For any $w$ with $\|w\|_H = 1$,

$$|w^\top T(\dot{\gamma},\dot{\gamma})| = |D^3F[\dot{\gamma},\dot{\gamma},w]| \le 2M_f\,\|\dot{\gamma}\|_H^2\,\|w\|_H = 2M_f\,\|\dot{\gamma}\|_H^2,$$

where the inequality is the polarized form of (10). Since $\|\dot{\gamma}(t)\|_{H(\gamma(t))} = \|v\|_{H_0}$ (constant speed on a geodesic), the bound follows. $\square$

To convert from the instantaneous norm $H(\gamma(t))$ to the base-point norm $H_0$, we use the Hessian comparison theorem: for $\beta = M_f\|\gamma(t) - s_0\|_{H_0} < 1$,

$$(1-\beta)^2\,H_0 \;\preceq\; H(\gamma(t)) \;\preceq\; (1-\beta)^{-2}\,H_0. \tag{12}$$

In particular, $\|w\|_{H_0} \le (1-\beta)^{-1}\|w\|_{H(\gamma)}$ for any vector $w$.

**Lemma 2** (Trajectory containment)**.** *If $M_f\delta < \frac{1}{4}$, then $\gamma(t)$ remains inside the ball $M_f\|\gamma(t) - s_0\|_{H_0} < \frac{1}{2}$ for $t \in [0,h]$. In particular, $\|\ddot{\gamma}(t)\|_{H_0} \le 2M_f\|v\|_{H_0}^2$.*

*Proof.* Define $\beta(t) = M_f\|\gamma(t) - s_0\|_{H_0}$. On any interval $[0,t_*]$ where $\beta < \frac{1}{2}$, Lemma 1 and (12) give

$$\|\ddot{\gamma}(\tau)\|_{H_0} \le \frac{M_f\|v\|_{H_0}^2}{1-\beta(\tau)} \le 2M_f\|v\|_{H_0}^2.$$

Integrating: $\|\dot{\gamma}(\tau) - v\|_{H_0} \le 2M_f\|v\|_{H_0}^2\,\tau$, so

$$\begin{aligned}\beta(t) = M_f\left\|\int_0^t \dot{\gamma}(\tau)\,d\tau - s_0\right\|_{H_0} &\le M_f\big(t\|v\|_{H_0} + M_f\|v\|_{H_0}^2 t^2\big)\\ &\le M_f\delta + (M_f\delta)^2 < \tfrac{1}{4} + \tfrac{1}{16} = \tfrac{5}{16} < \tfrac{1}{2}.\end{aligned}$$

The set $\{t : \beta(t) < \frac{1}{2}\}$ is open and contains 0; the bound shows it cannot have a first exit time in $[0,h]$. $\square$

### 10.2.2 Tangent line (order 1)

**Theorem 5** (Tangent line, general $M_f$)**.** *If $M_f\delta < \frac{1}{4}$, then*

$$\|\gamma(h) - (s_0 + hv)\|_{H_0} \le M_f\,\delta^2.$$

*Proof.* By Lemma 2, $\|\ddot{\gamma}(t)\|_{H_0} \le 2M_f\|v\|_{H_0}^2$ for $t \in [0,h]$. Since $\gamma(h) - s_0 - hv = \int_0^h (h-t)\,\ddot{\gamma}(t)\,dt$,

$$\|\gamma(h) - s_0 - hv\|_{H_0} \le 2M_f\|v\|_{H_0}^2 \cdot \frac{h^2}{2} = M_f\,\delta^2. \qquad \square$$

**Remark.** The factor of 2 (compared to the leading-order bound $\frac{1}{2}M_f\delta^2$ from Theorem 4) is the price of a rigorous global bound: Lemma 2 controls $\|\ddot{\gamma}\|_{H_0}$ uniformly over the trajectory, not just at $t = 0$. For small $M_f\delta$, the true error approaches $\frac{1}{2}M_f\delta^2$.

### 10.2.3 Legendre midpoint (order 2)

**Theorem 6** (Legendre midpoint, general $M_f$)**.** *If $M_f \delta < \frac{1}{4}$, then*

$$\|\gamma(h) - s_{\mathrm{Leg}}\|_{H_0} \leq C\, M_f^2\, \delta^3,$$

*where $C$ is a universal constant.*

*Proof.* Define the dual variable along the geodesic: $\lambda(t) = -\nabla F(\gamma(t))$, with $\dot\lambda = H(\gamma)\dot\gamma$.

*Step 1: bound $\ddot\lambda$.* Differentiating: $\ddot\lambda = D^3F[\dot\gamma, \dot\gamma, \cdot] + H\ddot\gamma = T(\dot\gamma, \dot\gamma) + H\ddot\gamma$. The geodesic equation gives $H\ddot\gamma = -\frac{1}{2}T$, so

$$\ddot\lambda = \tfrac{1}{2}T(\dot\gamma, \dot\gamma). \tag{13}$$

By the same argument as Lemma 1, $\|\ddot\lambda\|_{H^{-1}} = \frac{1}{2}\|T\|_{H^{-1}} \leq M_f\|v\|_{H_0}^2$.

*Step 2: bound $\dddot\lambda$.* Differentiating (13): $\dddot\lambda = D^4F[\dot\gamma, \dot\gamma, \dot\gamma, \cdot] + D^3F[\ddot\gamma, \dot\gamma, \cdot] + \text{sym}$. Differentiating the self-concordance inequality (10) gives $|D^4F[u, v, v, v]| \leq C_1 M_f^2 \|u\|_H \|v\|_H^3$ and the $D^3F[\ddot\gamma, \cdot, \cdot]$ terms are bounded using $\|\ddot\gamma\|_H \leq M_f\|v\|_H^2$ (Lemma 1). Together, in the $H_0^{-1}$ norm (using (12) with $\beta < \frac{1}{2}$):

$$\|\dddot\lambda(t)\|_{H_0^{-1}} \leq C_2 M_f^2 \|v\|_{H_0}^3 \tag{14}$$

for a universal constant $C_2$.

*Step 3: midpoint quadrature error.* The Legendre midpoint computes (Section 3.1):

$$\lambda_1^{\mathrm{Leg}} = -2\nabla F(s_0 + \tfrac{h}{2}v) + \nabla F(s_0) = \lambda_0 + h\dot\lambda_*,$$

where $\dot\lambda_* = H(s_0 + \frac{h}{2}v)\, v$ is the dual velocity evaluated at the *primal* midpoint $s_0 + \frac{h}{2}v$.

The true geodesic endpoint in dual coordinates is

$$\lambda(h) = \lambda_0 + \int_0^h \dot\lambda(t)\, dt.$$

The difference $\lambda_1^{\mathrm{Leg}} - \lambda(h)$ has two sources of error:

1. *Midpoint quadrature.* Even if we evaluated $\dot\lambda$ at the true geodesic midpoint $\gamma(h/2)$, the midpoint rule $\int_0^h f(t)\, dt \approx hf(h/2)$ has error $\frac{h^3}{24}\ddot f(\xi)$. Applied to $f = \dot\lambda$: error $= \frac{h^3}{24}\dddot\lambda(\xi)$, bounded by (14).

2. *Primal vs. geodesic midpoint.* The Legendre step evaluates at $s_0 + \frac{h}{2}v$ instead of $\gamma(h/2)$. These differ by $O(h^2)$ (Theorem 5 at half-step), so $H(s_0 + \frac{h}{2}v)v - H(\gamma(h/2))\dot\gamma(h/2) = O(M_f\|v\|_H^2 h)$. Multiplied by $h$: $O(M_f\delta^2 h) = O(M_f\delta^3/\|v\|_H)$. But this error enters at $O(h^3)$ because the midpoint evaluation is already correct to $O(h)$ in velocity.

Combining: $\|\lambda_1^{\mathrm{Leg}} - \lambda(h)\|_{H_0^{-1}} \leq C_3 M_f^2 \|v\|_{H_0}^3 h^3 = C_3 M_f^2 \delta^3$.

*Step 4: dual to primal.* For the barrier $F$, the gradient map $s \mapsto -\nabla F(s)$ has Jacobian $H(s)$. By the inverse function theorem, a dual error $\Delta\lambda$ in $H_0^{-1}$ norm produces a primal error $\|\Delta s\|_{H_0} \leq (1-\beta)^{-2}\|\Delta\lambda\|_{H_0^{-1}} \leq 4\|\Delta\lambda\|_{H_0^{-1}}$.

Therefore $\|\gamma(h) - s_{\mathrm{Leg}}\|_{H_0} \leq C\, M_f^2\, \delta^3$. □

#### 10.2.4 Yoshida-$2p$ (order $2p$)

**Theorem 7** (Yoshida composition, general $M_f$)**.** *If $M_f\delta < \frac{1}{4}$, then the Yoshida-$2p$ composition of Legendre midpoint steps satisfies*

$$\|\gamma(h) - s_{\mathrm{Y}_{2p}}\|_{H_0} \le C_p\, M_f^{2p}\, \delta^{2p+1}.$$

*Proof.* The Yoshida composition of order $2p$ is built from $2p-1$ Legendre substeps with coefficients $c_1, \ldots, c_{2p-1}$ summing to 1, chosen so that the local error expansion

$$s_{\mathrm{Leg}}(c_i h) = \gamma(c_i h) + c_i^3 h^3\, e_3 + c_i^5 h^5\, e_5 + \cdots$$

(where $e_k$ depends on $D^{k+1}F$ contracted with $v$) satisfies $\sum_i c_i^3 = 0$ (cancels $O(h^3)$), $\sum_i c_i^5 = 0$ (cancels $O(h^5)$), and so on through $\sum_i c_i^{2p-1} = 0$.

The leading surviving error has coefficient $\sum_i c_i^{2p+1}$ multiplied by $e_{2p+1}$, which involves $D^{2p+2}F$ contracted with $v$. By iterated differentiation of the self-concordance inequality (10), the $k$-th derivative of $F$ satisfies

$$|D^k F[v, \ldots, v]| \le C_k'\, M_f^{k-2}\, \|v\|_H^k$$

for universal constants $C_k'$ (the pattern: each differentiation of $|D^3F| \le 2M_f\|v\|_H^3$ introduces one factor of $M_f$ from the Hessian comparison).

The error term $e_{2p+1}$ involves $H^{-1}D^{2p+2}F[v^{2p+1}]$ (schematically), contributing $M_f^{2p}\|v\|_H^{2p+1}$. Multiplied by $h^{2p+1}$:

$$\|e_{2p+1}\|_{H_0} \cdot h^{2p+1} \le C_p\, M_f^{2p}\, \delta^{2p+1}.$$ □

**Remark.** The $M_f^{2p}$ scaling shows that higher-order methods are more sensitive to $M_f$ in the *coefficient*, but less sensitive in the *exponent* of $\delta$. For target error $\varepsilon$:

$$\delta_{\max} \propto \left(\frac{\varepsilon}{M_f^{2p}}\right)^{1/(2p+1)}.$$

This is the content of the step-size table below.

### 10.3 Step-size and work comparison

#### 10.3.1 Max step size for unit error

For target error $\varepsilon$, each method's max step size scales as $\delta_{\max} \propto (\varepsilon/M_f^{2p})^{1/(2p+1)}$:

| **Method** | **Error bound** | $\delta_{\max}$ **scaling** | $M_f = 1$ | $M_f = 10$ |
|---|---|---|---|---|
| Tangent line | $M_f\,\delta^2$ | $M_f^{-1/2}$ | 1.00 | 0.32 |
| Legendre midpoint | $CM_f^2\,\delta^3$ | $M_f^{-2/3}$ | $\sim 1$ | 0.22 |
| Yoshida-4 | $C_4M_f^4\,\delta^5$ | $M_f^{-4/5}$ | $\sim 1$ | 0.16 |
| Yoshida-6 | $C_6M_f^6\,\delta^7$ | $M_f^{-6/7}$ | $\sim 1$ | 0.12 |

**Observation.** The max step size for unit error actually *decreases* with method order when $M_f > 1$. The coefficient $M_f^{2p}$ grows faster than the exponent $\delta^{2p+1}$ can compensate. This is because the leading error term at order $2p$ involves the $(2p+2)$-th derivative of $F$, which self-concordance bounds as $O(M_f^{2p})$.

### 10.3.2 Where higher order helps: fixed step accuracy

The correct comparison is at a **fixed step size** $\delta$ (constrained to $M_f\delta < 1/4$ by the trajectory containment lemma). For example, at $\delta = 0.1/M_f$:

| **Method** | **Error at** $\delta = 0.1/M_f$ | **Ratio** |
|---|---|---|
| Tangent line | $M_f \cdot (0.1/M_f)^2 = 0.01/M_f$ | 1 |
| Legendre midpoint | $CM_f^2 \cdot (0.1/M_f)^3 = 10^{-3}C/M_f$ | $0.1C$ |
| Yoshida-4 | $C_4M_f^4 \cdot (0.1/M_f)^5 = 10^{-5}C_4/M_f$ | $10^{-3}C_4$ |
| Yoshida-6 | $C_6M_f^6 \cdot (0.1/M_f)^7 = 10^{-7}C_6/M_f$ | $10^{-5}C_6$ |

At any fixed fraction of the Dikin radius, higher-order methods give dramatically better accuracy. The error ratio is independent of $M_f$: each Yoshida level gains a factor of $\sim$100 for the same step size. This matches the 94,000× improvement observed in the exponential cone experiments (Section **??**).

### 10.3.3 Total work comparison

For a target per-step accuracy $\varepsilon$ and total path length $D$ (in Hessian norm), the total number of Hessian solves is (step count) × (solves per step):

| **Method** | **Solves/step** | $\delta(\varepsilon)$ | **Total solves** |
|---|---|---|---|
| Tangent | 1 | $(\varepsilon/M_f)^{1/2}$ | $D\,(M_f/\varepsilon)^{1/2}$ |
| Legendre | $\sim$3 | $(\varepsilon/CM_f^2)^{1/3}$ | $3D\,(CM_f^2/\varepsilon)^{1/3}$ |
| Yoshida-4 | 9 | $(\varepsilon/C_4M_f^4)^{1/5}$ | $9D\,(C_4M_f^4/\varepsilon)^{1/5}$ |

**Scaling with $\varepsilon$.** For fixed $M_f$, tangent scales as $\varepsilon^{-1/2}$, Legendre as $\varepsilon^{-1/3}$, Yoshida-4 as $\varepsilon^{-1/5}$. Higher order wins decisively when high per-step accuracy is needed.

**Scaling with $M_f$.** Tangent scales as $M_f^{1/2}$, Legendre as $M_f^{2/3}$, Yoshida-4 as $M_f^{4/5}$. Higher order has *worse* $M_f$ scaling. The $M_f^{2p}$ error coefficient (from higher derivatives of $F$) grows faster than the step-size exponent can absorb.

**Summary.** Higher-order Yoshida methods are most valuable when:

- The barrier is well-behaved ($M_f$ moderate), so the coefficient penalty is small,
- High per-step accuracy is needed (geodesic step must closely approximate the true geodesic),
- The overhead of multiple Hessian solves per step is acceptable.

For barriers with $M_f \gg 1$, higher order does not compensate for the poor regularity — the error coefficient $M_f^{2p}$ dominates. In this regime, the tangent-line step (with smaller steps) may be more efficient.

## 11 Application to interior-point methods

In conic optimization, the barrier function $F$ defines a Hessian manifold on the interior of the cone. The geodesic step keeps iterates strictly inside the cone (unlike a linear step which may exit). The Yoshida–Legendre method provides:

- **Cone-preserving steps** without computing third derivatives of the barrier,
- **Fourth-order accuracy** from three Newton solves per step (each on a system the size of the cone),
- **Generality**: works for any cone with a computable barrier gradient and Hessian (exponential cone, power cone, quantum entropy cone, etc.).

For the exponential cone (3-dimensional), the method converges in 4 factorizations on test problems, with the duality gap decreasing by a factor of $\sim 100$ per factorization.

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